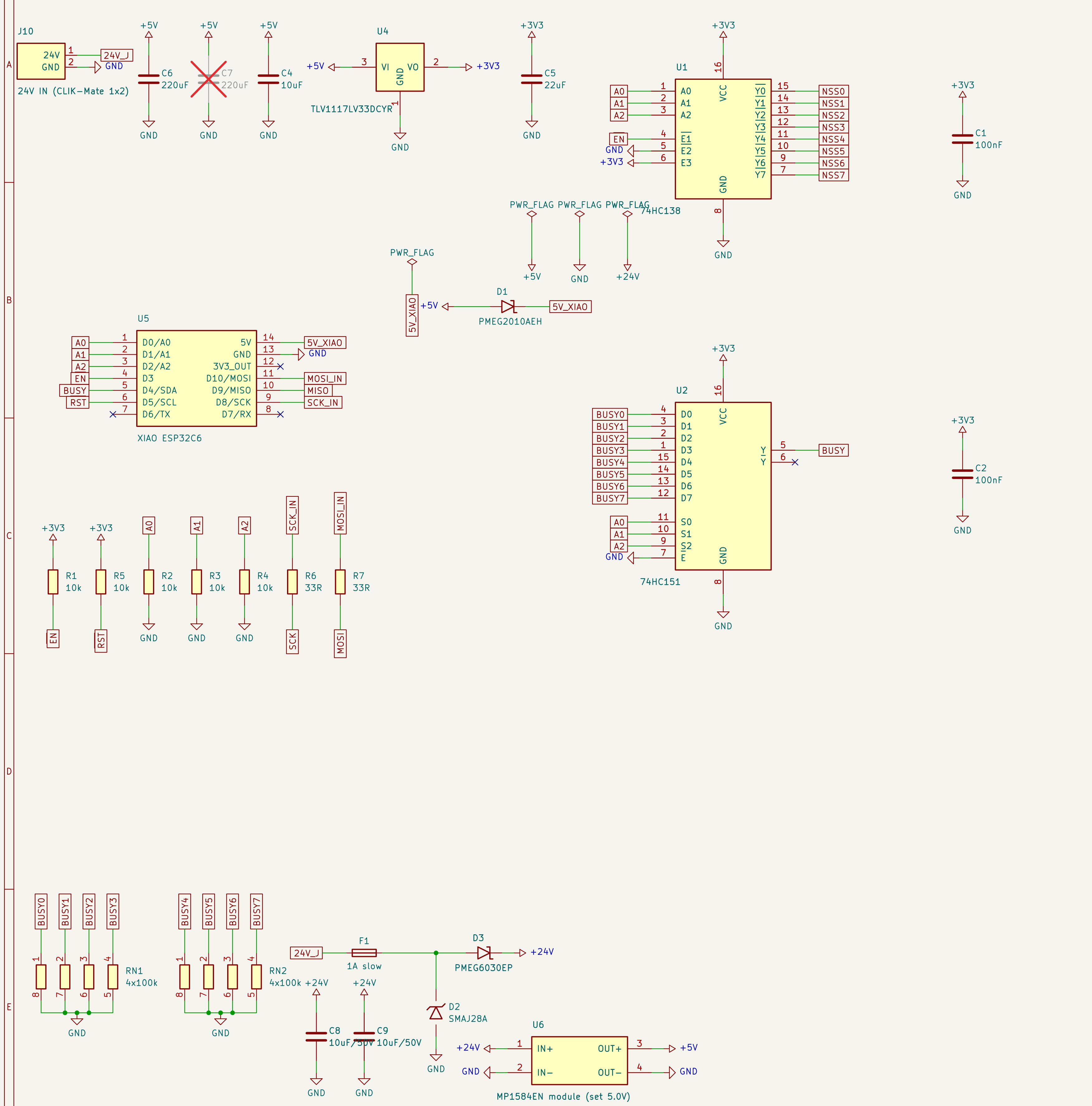

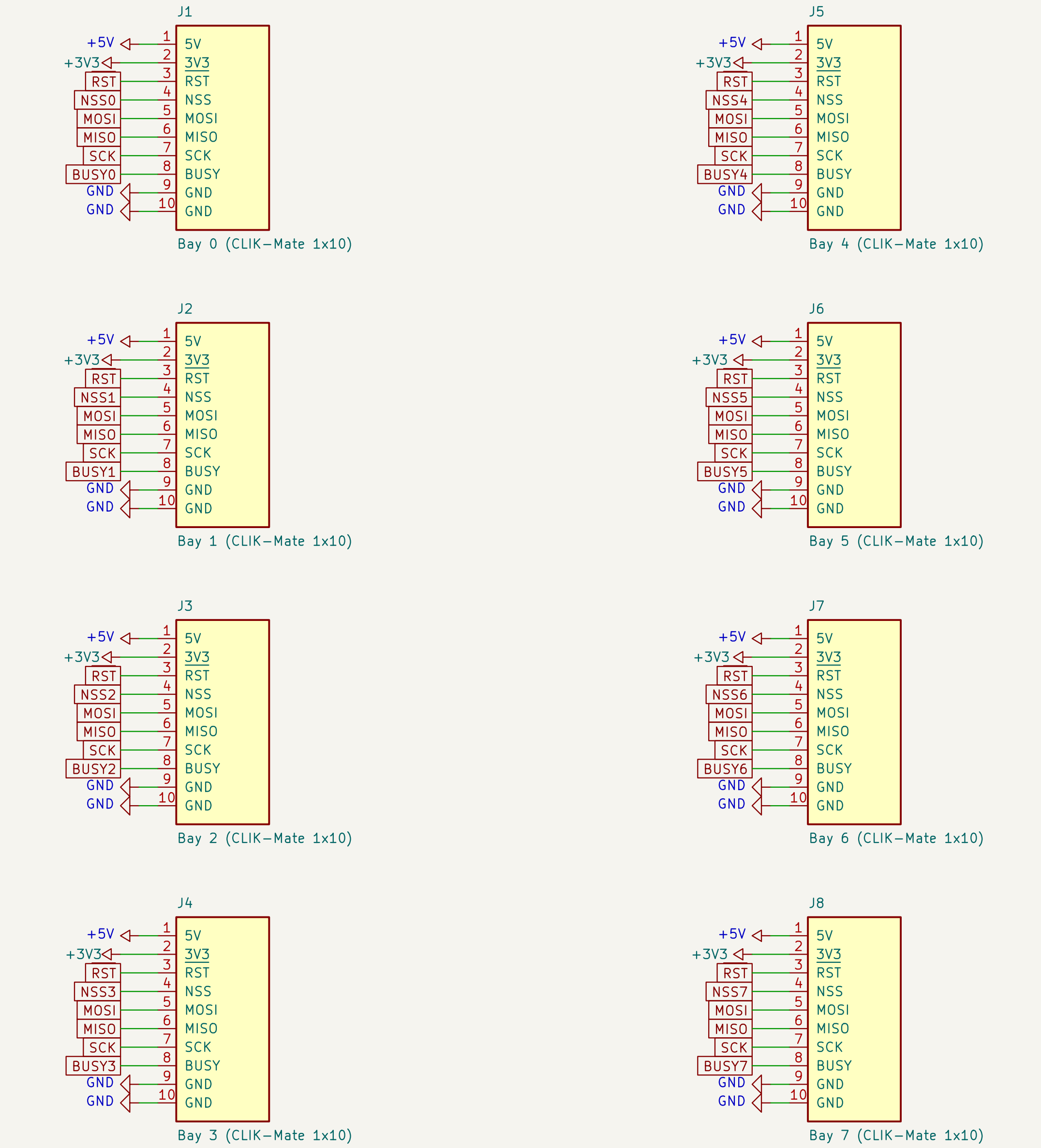

PN5180 8-BAY CARRIER – DESIGN NOTES

XIAO ESP32C6 pins: D8 SCK, D9 MISO, D10 MOSI, D3 $\overline{\text{EN}}$, D0 A0, D1 A1, D2 A2, D4 BUSY, D5 $\overline{\text{RST}}$; D6/D7 spare.

Bay select: A0..A2 -> 74HC138 address; $\overline{\text{EN}}$ (active low) = NSS of the addressed bay only.

BUSY = 74HC151 output for the addressed bay. Change A0..A2 only while $\overline{\text{EN}}$ is high.

Bays 4..7 (J5..J8) may be left unpopulated for a 4-bay build; RN1/RN2 keep unused BUSY inputs low.

Carrier 5V -> D1 Schottky -> XIAO 5V pin: the XIAO 5V pin is raw USB VBUS, D1 keeps the carrier from back-feeding USB.

Carrier 3V3 feeds bay pin 2 directly. Leave pin 2 out of the cable if your modules regulate 3V3 onboard.

Power: 24V from the printer PSU on J10 (~0.5 A). F1 1A -> D2 TVS -> D3 reverse Schottky -> C8/C9 -> U6 MP1584EN module -> 5V.

U6: MP1584EN mini module (28V max in) soldered flat on 4 pads. Set its pot to 5.00V BEFORE fitting; lock the pot with a dab of varnish.

Keep SPI <= 1 MHz with 8 bays on shared SCK/MOSI/MISO. Bay pinout matches the module header order.

All SMD except the XIAO sockets. Connectors: Molex CLIK-Mate 2.00mm RA receptacles (502494-xx70), housings 502439-xx00, terminals 502438.

Cables ~30 cm: R6/R7 (33R) source-terminate SCK/MOSI for the 8-way fan-out. SPI <= 1 MHz, ESP32 GPIO drive strength low.

U5 sits on two 1x7 female headers, 15.24 mm apart; antenna end (opposite USB) over the board edge, no copper under it.

Bay connector (J1..J8): 1=5V 2=3V3 3=$\overline{\text{RST}}$ 4=NSS 5=MOSI 6=MISO 7=SCK 8=BUSY 9=GND 10=GND (2nd return, twist with SCK).

U5 pins 1–7 = D0..D6 (USB end first), 8–11 = D7..D10, 12 = 3V3 out (NC), 13 = GND, 14 = 5V/VBUS (USB end)

Generated with gen_kicad.py
74HC138 chip-select decoder + 74HC151 BUSY mux, 5 ESP32 GPIOs for 8 readers
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Sheet: /
File: pn5180_carrier.kicad_sch

Title: PN5180 8-Bay NFC Reader Carrier

Size: A3 | Date: 2026-09-04 | **Rev: 1.3**
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